

Prathamesh Bhavsar

Dallas, TX (Open to Relocation) | prathameshbhavsar19@gmail.com | +1(940)279-9009 | GitHub | Website | LinkedIn

Summary

AI/ML Engineer with 4 years of experience spanning traditional applied ML and modern GenAI systems, from predictive modeling and ML pipelines to LLM-based and RAG solutions in production. Experienced in delivering AI/ML systems for regulated, high-stakes environments such as healthcare and energy. Skilled in owning the full lifecycle: requirements, development, deployment, monitoring, and stakeholder communication.

Skills

- **Programming & Databases** - Python, SQL, Spark, Snowflake, MongoDB, NumPy, Pandas
- **AI/ML & GenAI** - TensorFlow, PyTorch, LangChain, GPT-4, BioBERT, RAG, Vector Search, Prompt Engineering
- **Cloud & MLOps** - AWS, SageMaker, Bedrock, EC2, S3, Step Functions, MLflow, Airflow, Docker, CI/CD

Experience

Data and AI Co-Op, Tenet Healthcare - Dallas, TX Aug 2025 – May 2026

- Designed and delivered a production-grade **GenAI medical coding solution** using **LangChain and GPT-4**, translating stakeholder requirements into scalable AI workflows; improved coding accuracy from **78% to 89%** and reduced manual effort from **100 hours to 45 hours**.
- Built an end-to-end **clinical trial matching pipeline** using **BioBERT, graph-based retrieval, and ML workflows**, improving screening efficiency and increasing enrollment from **100 to 142 patients**.
- Operationalized a real-time **AI decision support system** with deployment, monitoring, and stakeholder-facing insights across **8 ICUs**.
- Collaborated with healthcare, data, and engineering teams to align **AI/ML solutions** with business objectives, compliance-sensitive workflows, and production delivery requirements.

Software Development Engineer (Machine Learning), EnelX - Mumbai, India Jul 2021 – Aug 2024

- Developed and deployed scalable **machine learning services** for predictive maintenance, supporting high-throughput data processing and reducing outages from **100 incidents to 60 incidents**.
- Designed an **AI-driven energy analytics platform** for business stakeholders, transforming enterprise data into actionable insights and reducing average energy costs from **\$100K to \$76K** across **300+ facilities**.
- Built distributed **data and ML pipelines** integrating streaming workflows and knowledge graphs, improving reliability and reducing root cause identification time from **2 hours to 35 minutes**.
- Created an optimization engine for **EV fleet operations**, operationalizing predictive models to improve cost efficiency and increase uptime from **90% to 97%** across **800+ vehicles**.
- Led cross-functional data analysis across **30+ suppliers**, improving data quality, reducing duplicate overhead from **12% to 5%**, and saving effort from **100 hours to 50 hours monthly**.
- Designed reusable backend services and **MLOps-ready data pipelines** processing millions of records daily, improving system efficiency from **70% to 90%**.

Education

University of Texas at Dallas, MS in Computer Engineering Aug 2024 – May 2026

University of Mumbai, Bachelor of Engineering in Information Technology Jul 2017 – Jun 2021

Projects

PitWall — Production RAG API over F1 FIA Regulations (FastAPI, Qdrant, GPT-4o, Langfuse) GitHub Link

- Built and deployed a production **hybrid RAG API** (dense + BM25 + RRF retrieval, GPT-4o generation) over FIA regulation text, achieving **0.84 Context Precision** and **0.87 Faithfulness** on RAGAS evaluation.
- Implemented a **CI/CD pipeline** with automated RAGAS evaluation gates and full request-level **observability** (cost, latency, token tracing via Langfuse), load-tested to 150 concurrent users on Azure Container Apps.

AgentToyota — Smart Vehicle Financing AI Agent (FastAPI, React, LangChain) GitHub Link

- Developed a full-stack **AI agent workflow** using FastAPI, React, and LangChain, enabling scalable decision automation and reducing decision time by **45%**.

StreamFraud: Real-Time Credit Card Fraud Detection (Kafka, Docker, Isolation Forest) GitHub Link

- Built a containerized **real-time ML pipeline** using Kafka and Docker for scalable anomaly detection, achieving **99.79% accuracy**.